

AP Statistics FRQ Practice Exam 07

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AP Statistics FRQ Practice Exam 7

APSTAT-FRQ7-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 4, Inference for Quantitative Data: Means

FRQ Type: Formulate Questions and Collect Data

Prompt:

A greenhouse manager wants to study how light spectrum and watering schedule affect the dry leaf mass of basil plants after 6 weeks. From a batch of 600 similar seedlings, 80 plants are selected at random. The two factors and their levels are

Factor 1, light spectrum: Full-spectrum or Blue-enriched

Factor 2, watering schedule: Daily or Alternate-day

The 80 plants are randomly assigned in equal numbers to the four treatment combinations. Mean dry leaf mass for each group is shown.

Light spectrum	Daily watering	Alternate-day watering
Full-spectrum	42 g	34 g
Blue-enriched	48 g	38 g

- Identify the population, sample, and experimental units.
- Identify the factors, levels, treatments, and response variable. State a statistical question about whether the effect of light spectrum depends on watering schedule.
- Describe a valid random-assignment procedure that creates four groups of 20 plants. Explain the purpose of random assignment.
- Calculate the average effect of changing from Full-spectrum to Blue-enriched light across the two watering schedules. Also calculate the average effect of changing from Alternate-day to Daily watering across the two light spectra. Interpret both effects.
- Use the four group means to assess whether there is evidence of an interaction between light spectrum and watering schedule. State the causal and generalization conclusions that the design can support, without performing a significance test.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as the batch of 600 similar basil seedlings and the sample as the 80 seedlings selected at random.
- Point 2: Identifies the individual basil plants as the experimental units.

(b) 2 points

- Point 1: Identifies light spectrum and watering schedule as the factors, their two stated levels, the four combinations as treatments, and dry leaf mass after 6 weeks in grams as the response.
- Point 2: States a question such as, “Does the effect of Blue-enriched versus Full-spectrum light on mean dry leaf mass differ between Daily and Alternate-day watering?”

(c) 2 points

- Point 1: Describes labeling all 80 plants and using a random process to assign 20 plants to each of the four treatment combinations.
- Point 2: Explains that random assignment tends to balance other plant characteristics and growing-location effects across treatments, supporting causal comparisons.

(d) 2 points

- Point 1: Calculates the average light-spectrum effect as $[(48 + 38)/2] - [(42 + 34)/2] = 43 - 38 = 5$ g and interprets Blue-enriched light as producing 5 g greater mean dry leaf mass, averaged over watering schedules.
- Point 2: Calculates the average watering effect as $[(42 + 48)/2] - [(34 + 38)/2] = 45 - 36 = 9$ g and interprets Daily watering as producing 9 g greater mean dry leaf mass, averaged over light spectra.

(e) 2 points

- Point 1: Compares the light effect under Daily watering, $48 - 42 = 6$ g, with the light effect under Alternate-day watering, $38 - 34 = 4$ g; the 2 g difference suggests a small possible interaction, but the means alone do not establish whether it is more than experimental variation.
- Point 2: States that random assignment supports causal conclusions for the experimental plants and random selection supports cautious generalization to the original batch of 600 similar seedlings, but not automatically to other basil varieties or growing conditions.

Model Guidance:

This is a 2×2 factorial experiment. The four treatments are the combinations of two light spectra and two watering schedules. Randomly allocate the 80 selected seedlings so that each treatment receives 20 plants; other greenhouse conditions should be held as similar as possible.

Averaging over the other factor gives a 5 g increase for Blue-enriched light and a 9 g increase for Daily watering. To examine interaction, compare one factor's effect at each level of the other factor. The light effect is 6 g under Daily watering and 4 g under Alternate-day watering. Those effects are not identical, but group means alone do not measure whether the 2 g difference is statistically meaningful. Random selection and random assignment support different conclusions: generalization to the seedling batch and causation among the assigned treatments, respectively.

Expected Points:

10

APSTAT-FRQ7-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

The diameters of mature blueberries from a harvest containing more than 3,600 berries are adequately modeled by a normal distribution with mean $\mu = 16.0$ millimeters and standard deviation $\sigma = 2.4$ millimeters. Let X be the diameter of one randomly selected berry. Let $\bar{x}_{36}$ be the mean diameter of an SRS of 36 berries from the harvest.

- (a) Describe the distributions of X and $\bar{x}_{36}$, including shape, mean, and standard deviation.
- (b) Calculate $P(X < 15.2)$. Interpret the probability in context.
- (c) Calculate $P(\bar{x}_{36} < 15.2)$. Interpret the probability in context.
- (d) Explain why the probabilities in parts (b) and (c) differ substantially even though X and $\bar{x}_{36}$ have the same mean.
- (e) Let $\bar{x}_9$ be the mean diameter of an SRS of 9 berries. Calculate $P(\bar{x}_9 < 15.2)$ and explain why the result lies between the probabilities in parts (b) and (c).

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: States $X \sim \text{Normal}(16.0, 2.4)$, using 2.4 as the standard deviation.
- Point 2: States that $\bar{x}_{36}$ is normal with $\mu_{\bar{x}} = 16.0$ mm and $\sigma_{\bar{x}} = 2.4/\sqrt{36} = 0.4$ mm.

(b) 2 points

- Point 1: Calculates $z = (15.2 - 16.0)/2.4 \approx -0.333$ and $P(X < 15.2) \approx 0.3694$.
- Point 2: Interprets the result: about 36.9% of individual mature blueberries from the harvest are expected to have diameters below 15.2 mm.

(c) 2 points

- Point 1: Calculates $z = (15.2 - 16.0)/0.4 = -2.00$ and $P(\bar{x}_{36} < 15.2) \approx 0.0228$.
- Point 2: Interprets the result: about 2.28% of SRSs of 36 berries are expected to have a sample mean diameter below 15.2 mm.

(d) 2 points

- Point 1: Explains that averaging 36 independent observations reduces the standard deviation from 2.4 mm to 0.4 mm.
- Point 2: Connects the smaller spread to the cutoff: 15.2 mm is only 0.333 standard deviation below the individual mean but 2 standard deviations below the sample-mean center.

(e) 2 points

- Point 1: Calculates $\sigma_{\bar{x}9} = 2.4/\sqrt{9} = 0.8$ mm, $z = (15.2 - 16.0)/0.8 = -1.00$, and $P(\bar{x}_9 < 15.2) \approx 0.1587$.
- Point 2: Explains that averaging 9 berries produces less variability than one berry but more variability than an average of 36 berries, so its lower-tail probability is intermediate.

Model Guidance:

The individual distribution is Normal with center 16.0 mm and spread 2.4 mm. Because the source population is normal, each sample-mean distribution is also normal. Its center stays at 16.0 mm, while its standard deviation becomes $\sigma/\sqrt{n}$.

For an individual berry, 15.2 mm is about 0.333 standard deviation below the mean, giving probability 0.3694. For a mean of 36, the standard deviation is 0.4 mm, making the same cutoff 2 standard deviations below the center and reducing the probability to 0.0228. With $n = 9$, the standard deviation is 0.8 mm and the probability is 0.1587. Larger random samples produce more tightly clustered sample means without changing their center.

Expected Points:

10

APSTAT-FRQ7-003

Section: FRQ

Type: frq

Primary Unit: Unit 3, Inference for Categorical Data: Proportions

Secondary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A bicycle-sharing service claims that fewer than 4% of helmets returned during a month fail the service's safety inspection. From more than 10,000 helmet returns that month, a safety auditor takes a simple random sample of 500 helmets. Twelve sampled helmets fail inspection.

At the $\alpha = 0.05$ significance level, test the service's claim.

- Define the population parameter and state the hypotheses.
- Identify the appropriate inference procedure and check its conditions.
- Calculate the sample proportion, null standard error, test statistic, and p-value.
- State a conclusion in context using the $\alpha = 0.05$ significance level.
- Explain why statistical significance alone does not determine whether the estimated reduction below 4% is practically important.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines p as the true proportion of all helmets returned to the service during the month that would fail the safety inspection.
- Point 2: States $H_0: p = 0.04$ and $H_a: p < 0.04$.

(b) 3 points

- Point 1: Selects a one-sample z test for a population proportion.
- Point 2: Cites the SRS and verifies the 10% condition because 500 is less than 10% of more than 10,000 returns.
- Point 3: Checks the large-counts condition under H_0 : $np_0 = 500(0.04) = 20$ and $n(1 - p_0) = 480$, both at least 10.

(c) 3 points

- Point 1: Calculates $\hat{p} = 12/500 = 0.024$ and $SE_0 = \sqrt{[0.04(0.96)/500]} \approx 0.008764$.
- Point 2: Calculates $z = (0.024 - 0.04)/0.008764 \approx -1.83$.
- Point 3: Calculates the left-tail p -value $P(Z \leq -1.83) \approx 0.0339$.

(d) 1 point

- Point 1: Because $0.0339 < 0.05$, rejects H_0 and concludes that the sample provides convincing evidence that fewer than 4% of all helmets returned during the month would fail inspection.

(e) 1 point

- Point 1: Explains that practical importance depends on consequences, costs, and safety standards; a statistically detectable sample difference of $0.04 - 0.024 = 0.016$, or 1.6 percentage points, is not automatically large or operationally meaningful.

Model Guidance:

The claim concerns one population proportion, so use a one-proportion z test with a left-tailed alternative. Under H_0 , the standard error must use $p_0 = 0.04$. The random sample, 10% condition, and null expected counts of 20 failures and 480 passes justify the normal approximation.

The sample failure proportion is 0.024. Its null-standardized test statistic is approximately -1.83 , producing $p \approx 0.0339$. Because the p -value is below 0.05, the data support the service's claim for that month's returns. This statistical conclusion does not decide whether a 1.6-percentage-point difference matters enough to change policy; that judgment requires information about injury risk, replacement cost, and inspection standards.

Expected Points:

10

APSTAT-FRQ7-004

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A records center takes a simple random sample of 8 completed cataloging projects from 160 projects. For each project, x is the number of folders in an archival box and y is the cataloging time in hours.

Project	1	2	3	4	5	6	7	8
Folders, x	20	24	28	32	36	40	44	48
Cataloging time, y	17	21	25	29	30	32	32	30

The least-squares regression line is $\hat{y} = 10 + 0.5x$, with $r = 0.890$ and $r^2 = 0.792$.

- (a) Calculate the fitted values and residuals for Projects 1 and 4. Interpret the residual for Project 4.
- (b) Calculate the residuals for the remaining six projects. Verify that the residuals sum to 0 and explain why a residual sum of 0 does not show that the model predicts every project accurately.
- (c) Calculate the sum of squared errors, $SSE = \sum(y - \hat{y})^2$, and the residual standard deviation $s = \sqrt{[SSE/(n - 2)]}$. Interpret s in context.
- (d) On paper, sketch a dotplot of the eight residuals. Label the axis and describe the center, spread, and any notable features of the residual distribution as one-variable data.
- (e) A manager claims, "The model is highly accurate because its mean residual is 0, so no prediction misses the observed cataloging time by more than 3 hours." Evaluate the entire claim using the residuals and the residual standard deviation.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: For Project 1, calculates $\hat{y} = 10 + 0.5(20) = 20$ and residual $17 - 20 = -3$; for Project 4, calculates $\hat{y} = 10 + 0.5(32) = 26$ and residual $29 - 26 = 3$.
- Point 2: Interprets the Project 4 residual: the project took 3 hours longer than the model predicted for a 32-folder box.

(b) 2 points

- Point 1: Calculates the complete residual sequence as $-3, -1, 1, 3, 2, 2, 0, -4$.
- Point 2: Verifies that the sum is 0 and explains that zero mean residual is an automatic least-squares property with an intercept; positive and negative errors can cancel even when individual errors are substantial.

(c) 2 points

- Point 1: Calculates $SSE = 9 + 1 + 1 + 9 + 4 + 4 + 0 + 16 = 44$ square hours and $s = \sqrt{[44/(8 - 2)]} \approx 2.71$ hours.
- Point 2: Interprets 2.71 hours as the typical size of the vertical prediction errors when the line predicts cataloging time from folder count for projects like those sampled.

(d) 2 points

- Point 1: Draws a correctly labeled dotplot with residuals at $-4, -3, -1, 0, 1, 2, 2$, and 3.

· Point 2: Describes the distribution as centered at or near 0, ranging from -4 to 3 hours, with two residuals at 2 and no point clearly isolated far from all others.

(e) 2 points

· Point 1: Explains that mean residual 0 follows from fitting a least-squares line with an intercept and therefore is not, by itself, evidence of high predictive accuracy.

· Point 2: Rejects the “no miss greater than 3 hours” statement because Project 8 has residual -4 hours, and cites $s \approx 2.71$ hours as evidence that typical errors are not negligible relative to the claimed bound.

Model Guidance:

Apply residual = observed – predicted. The fitted values increase from 20 to 34 hours, producing residuals -3, -1, 1, 3, 2, 2, 0, and -4. A positive residual means the actual project took longer than predicted; a negative residual means it took less time.

The residuals sum to 0 because the fitted least-squares line includes an intercept, not because predictions are perfect. Squaring and summing prevents cancellation and gives $SSE = 44$. Dividing by $n - 2 = 6$ and taking the square root gives $s \approx 2.71$ hours, a measure of typical vertical error. The residual dotplot is centered near 0 but spans 7 hours. In particular, the -4-hour residual directly contradicts the manager's claimed maximum error.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ7

· APSTAT-FRQ7-001: Unit 1 primary; Unit 4 secondary; 2×2 factorial basil-seedling experiment with 80 randomly selected from a batch of 600 and randomly assigned 20 per combination of Full-spectrum/Blue-enriched light and Daily/Alternate-day watering; group mean dry leaf masses 42, 34, 48, and 38 g; identify population/sample/units/factors/levels/treatments/response, describe equal random assignment, calculate 5 g light and 9 g watering average effects, compare conditional light effects of 6 and 4 g for possible interaction, and separate causal from population scope; targets ignoring interaction and confusing random selection with random assignment.

· APSTAT-FRQ7-002: Unit 2 primary; Unit 1 secondary; normal blueberry-diameter model with $\mu = 16.0$ mm and $\sigma = 2.4$ mm; compare one berry with SRS means for $n = 36$ and $n = 9$ at the same 15.2 mm cutoff, using standard deviations 2.4, 0.4, and 0.8 and probabilities 0.3694, 0.0228, and 0.1587; explain unchanged center and shrinking sample-mean variability; targets using σ instead of $\sigma/\sqrt{n}$ and confusing an individual probability with a sample-mean probability.

· APSTAT-FRQ7-003: Unit 3 primary; Unit 2 secondary; left-tailed one-proportion z test of a bicycle-sharing service's claim that monthly helmet inspection failure proportion is below 0.04, using an SRS of 500 from more than 10,000 returns with 12 failures; state $H_0: p = 0.04$ and $H_a: p < 0.04$, check SRS/10%/null large counts, calculate $\hat{p} = 0.024$, $SE_0 \approx 0.008764$, $z \approx -1.83$, $p \approx 0.0339$, reject at $\alpha = 0.05$, and distinguish statistical from practical significance; targets using $\hat{p}$ in the null standard error and treating significance as operational importance.

· APSTAT-FRQ7-004: Unit 5 primary; Unit 1 secondary; SRS of eight archival-box cataloging projects with folders $x = 20-48$, times $y = 17, 21, 25, 29, 30, 32, 32, 30$, and line $\hat{y} = 10 + 0.5x$ with $r = 0.890$ and $r^2 = 0.792$; compute and interpret all residuals, verify their zero sum, calculate $SSE = 44$ and residual $SD \approx 2.71$ hours, sketch and describe a residual dotplot as one-variable data, and refute an accuracy claim using the -4 residual; targets treating residual cancellation as perfect prediction and confusing zero mean error with a maximum-error guarantee.